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----- Active space -----

STRUCTURE FUNCTION RELATIONSHIP IN LU

Conducting Airways

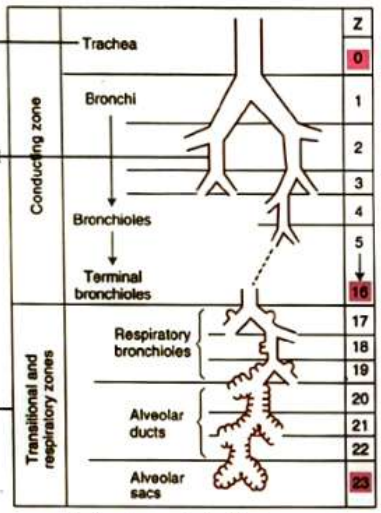
00:07:16

EPITHELIUM

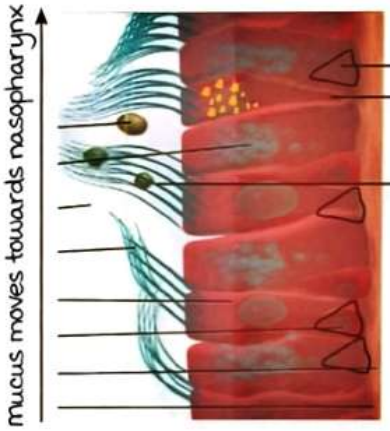
Dead space no gas exchange

Medium sized bronchi : Offers max. resistance to airflow

Gas exchange occurs



Weibel model of airway (23 airway generations)



- Basal cell
- Clara cell (Club cell)

Stem cells

Ciliary movement :

- Driven by protein : Dynein.
- Synchronized movement : metachronism.
- Functions : Clearing sputum.

Psuedostratified columnar epithelium

IMMOTILE CILIA SYNDROMES

1. Cystic fibrosis :

Pathophysiology :

Cystic fibrosis transmembrane conductance regulator (CFTR) gene mutation

↓

Defective Cl^- opening

↓

Thick sputum

Rx : Ivacaftor → Opens Cl^- channel.

2. Kartagener syndrome :

Pathophysiology : D/t absent dynein.

Clinical features :

1. Bronchiectasis
2. Sinusitis
3. Situs inversus

+ Asthenospermia : Immotile sperm

Note :

Cilia : Responsible for rotation of internal organs during development.

Dynein : Protein responsible for movement of → Cilia, Sperm, Fallopian tube



Dextrocardia

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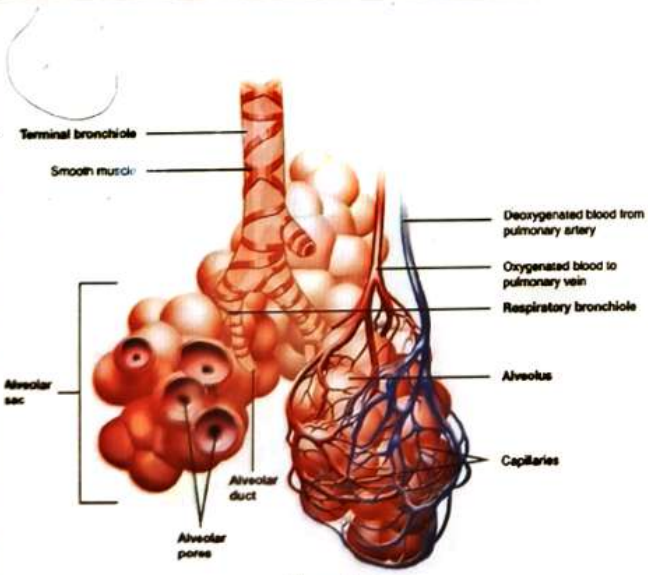
SMOOTH MUSCLES

----- Active space -----

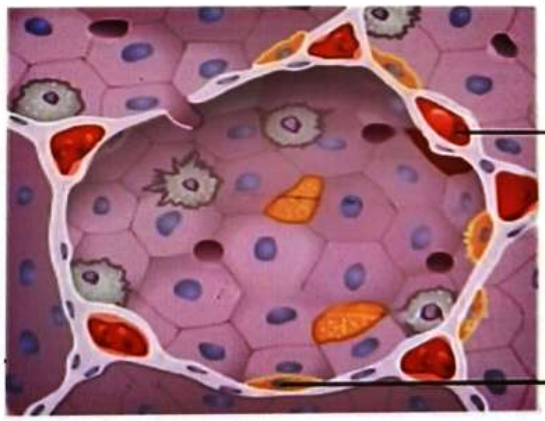
Features	Bronchoconstriction	Bronchodilation
Image		
Resistance	↑	↓
Airflow	↓	↑
Autonomic nervous system activity	Parasympathetic	Sympathetic (β2 receptor)
mediators	• Leukotriene (most potent). • Acetylcholine. • Histamine.	• Salbutamol (β2 agonist). • Nitric oxide. • Prostaglandin.
Applied aspect	Exaggerated bronchoconstriction ↓ Asthma	β2 agonists used in Rx of asthma

Alveolar Airways

00:25:10



Alveoli



- Large, flat.
- ↑ surface area.
- ↓ in number.

Pneumocytes

- Small.
- ↓ surface area.
- ↑ in number.
- Function :
  - Surfactant production.
  - Transforms into stem cell for regeneration.

Note :

Stem cells in airway :

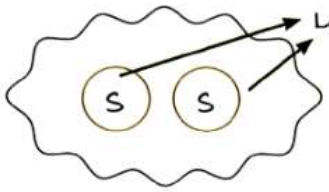
1. Basal cell.
2. Clara cell (club cell).
3. Type 2 pneumocyte.

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SURFACTANT

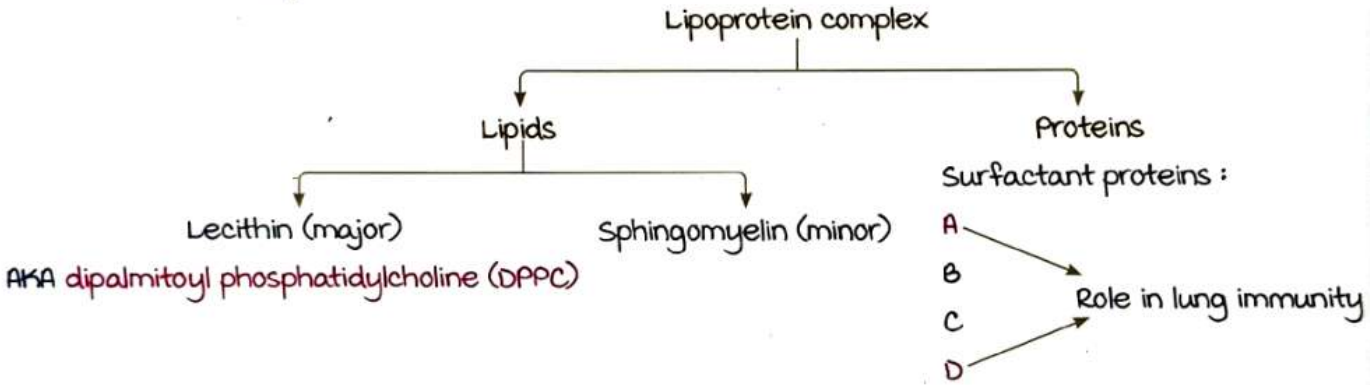
Production :



Type a pneumocyte

- Appears around 28–32 weeks of gestation.

Composition :



Functions :

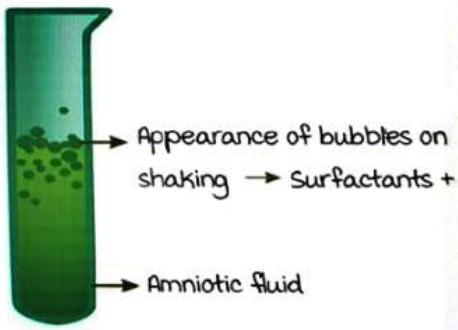
1. Fetal lung maturity :

   Tests :

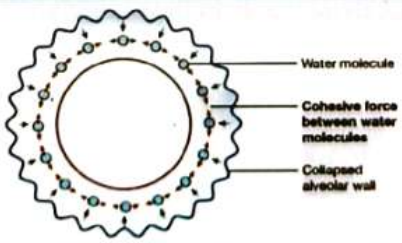
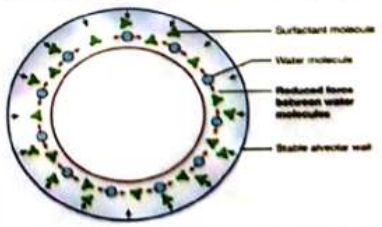
   a. Lamellar body counting.

   b. Lecithin & sphingomyelin ratio (L/S) : ≥ 2 → mature fetal lungs.

   c. Shake test/bubble test/clements test.

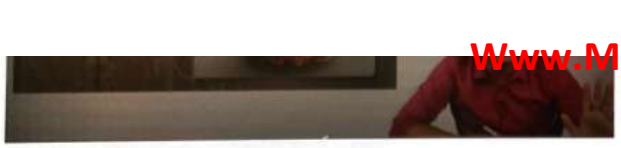


2. To prevent alveolar collapse :

Features	Alveoli without surfactant	Alveoli with surfactant
		
Surface tension (T)	↑	↓
Radius (r)	↓	↑
Area (πr^2)	↓	↑
Collapsing pressure (P) (Laplace law : $P = 2T/r$)	↑ (Alveoli collapse)	↓ (Alveoli doesn't collapse)

Feedback

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Applied aspect :

Respiratory distress syndrome (RDS) :

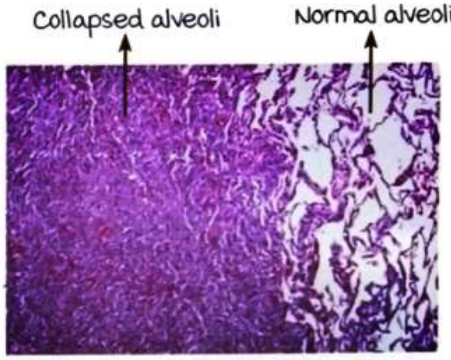
AKA hyaline membrane disease (HMD).

- Surfactant deficiency → Alveoli collapse.
- Rx :

  Surfactant drugs :

  a. Beractant.

  b. Lucinactant.



RDS : Histopathology

Regulation of surfactant production :

Factors

Stimulating (Physiological)

1. Steroid : Cortisol
   - Applied aspect : Used in preterm babies for lung maturation.
   - Eg :
     - Betamethasone.
     - Dexamethasone.
2. Thyroid hormones : T3, T4.

Inhibiting (Pathological)

1. High levels of insulin (Inhibits cortisol).
   - Applied aspect :

     Infants of diabetic mothers

     ↓

     Surfactant deficiency

     ↓

     RDS/HMD
2. Cigarette smoking.
3. O_2 toxicity.
4. Occlusion of main bronchus.
5. Occlusion of one pulmonary artery.

Feedback

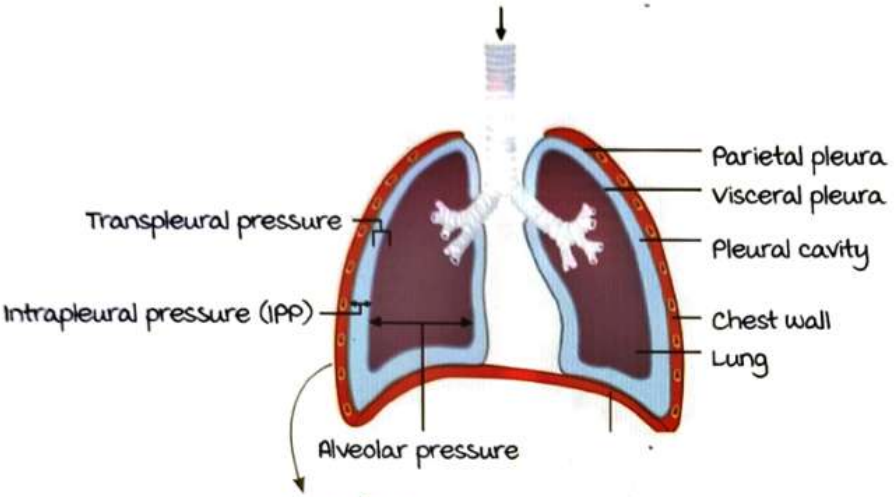
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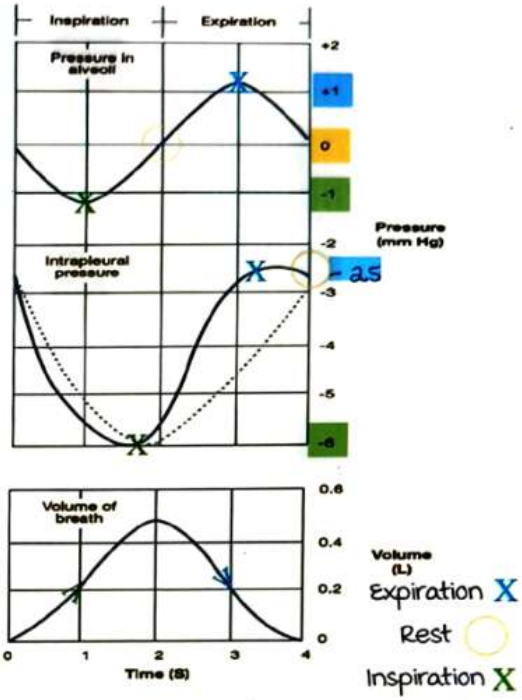
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MECHANICS OF BREATHING

Pressures in the Lung

00:02:36





Boyle's law : Pressure $\propto \frac{1}{\text{Volume (tidal volume)}}$

TYPES OF PRESSURE :

1. Intrapleural pressure (IPP) :

- At rest : Negative (D/t vacuum created by pleural stretch).
- Inspiration : more negative.
- Expiration : Same as rest.
- Forceful expiration : Positive.

2. Alveolar pressure (AP) :

At rest : Zero.

Inspiration : Negative.

Expiration : Positive.

3. Transpulmonary pressure = AP − IPP :

- At rest : Positive.
- Keeps alveoli distended.

Note :

- 1 mmHg = 1.36 cm H_2O.
- muscles of respiration :

Inspiration :

- Primary muscle : Diaphragm.
- Accessory muscle :
  - External intercostal muscle.
  - Scalene muscle.
  - Sternocleidomastoid muscle.

Expiration :

- Normal : Passive process.
- Forceful : Active process.
  - Rectus abdominus.
  - Transverse abdominis.
  - Internal intercostal muscle.

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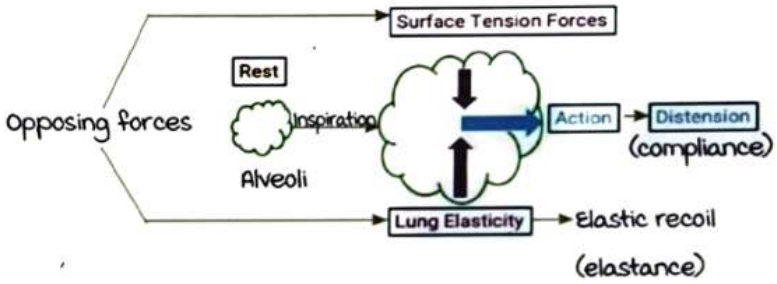
Lung compliance

00:19:30 ----- Active space -----

Lung Compliance : $\frac{\text{Change in volume}}{\text{Change in pressure}}$

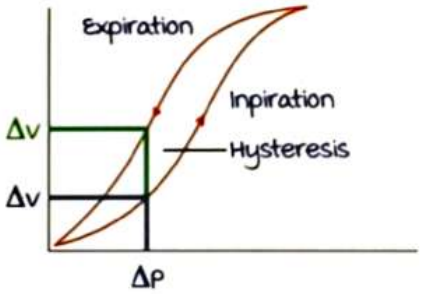
$C = \Delta V/\Delta P$ = 200 ml/cm H_2O (Normal).

Compliance $\propto \frac{1}{\text{Elastance}}$

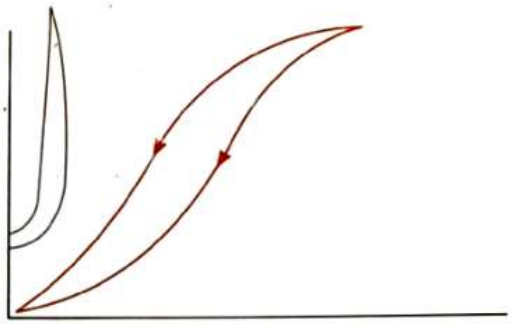


Hysteresis :

- Difference b/w inspiratory & expiratory compliance of lung.
  - Compliance greatest during expiration.
- Occurs d/t :
  1. Surface tension forces :
     Air filled vs saline filled lung :



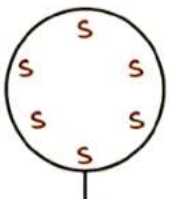
Saline filled → No air fluid interface



Air filled lung
↓
Air fluid interface
↓
↑ surface tension
↓
↑ hysteresis/↑ Compliance

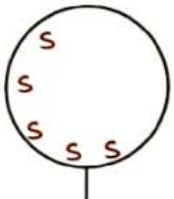
2. Surfactant :

Inspiration



Widely placed surfactant
↓
Less reduction in surface tension
↓
Less compliant

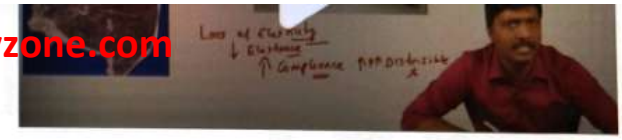
Expiration



Closely placed surfactant
↓
more reduction in surface tension
↓
more compliant

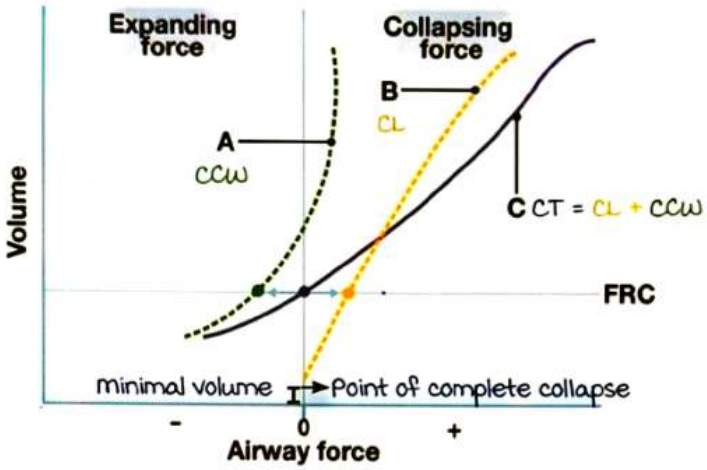
Feedback

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Compliance curves :



CCW : Compliance of chest wall.

CL : Compliance of lung.

CT : Total compliance.

FRC : Functional residual capacity (Point of equilibrium between expanding & collapsing forces).

Applied aspect : Minimum volume seen in pneumothorax.

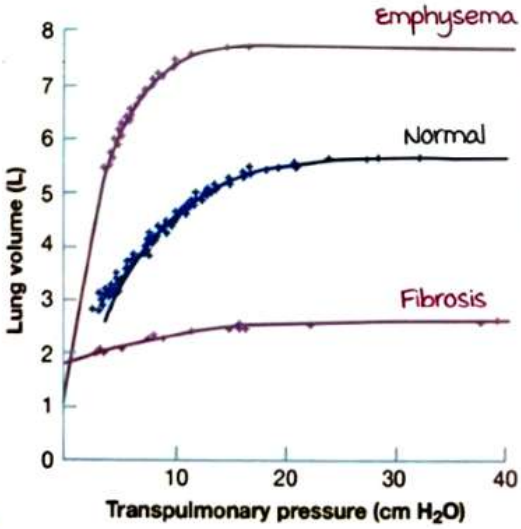
Types of Compliance :

1. Static :
During breath holding.

2. Dynamic :
During normal breathing.
- D/t impedance (Resistance to air flow).

3. Specific :
$= \frac{\text{Compliance}}{\text{FRC}}$

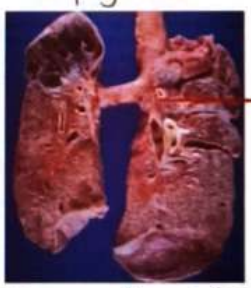
Applied aspect :





Emphysema

↑ compliance (D/t loss of elastance)



Pulmonary fibrosis/ILD

↓ compliance (D/t stiff lung)



Hyperinflation of lung

Feedback

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Work of breathing

00:46:16 ----- Active space -----

Work = Force x displacement = 0.5 kg/m/min (By lung).

- Pressure changes
- Volume changes

Work done (100%)

Overcome

- 65 % work-Elastic forces
  - 2/3rd Surface tension forces
  - 1/3rd Tissue elasticity
- 35 % work-Resistance
  - 28 % Airway resistance
  - 7 % Lung tissue resistance

Feedback